NPLify · P0 Technical Baseline

Ledger Patterns — a Reading Companion

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11 October 2026 · Confidential

Companion to the Ledger Posting Design & Chart of Accounts: the same 61 worked transactions, each with a plain-language explanation beside it.

How to read this pack

Four map pages are now part of the pack as documents of their own, and are the place to start: the Top-Down Map (from the business, through the data model and the deal states, to the scenarios and the reading order), the Deal Flow on one page (Quote, Collect, Convert, Settle, with the roles and the money path), the Workflow Tree (8 classes, 27 sub-classes, 92 procedures, each naming who initiates, who approves, the codes it is written under and the records it touches) and the Workflow Tree on one page. Read them in that order, then this document. Every worked transaction (T), acceptance vector (L, B), walkthrough (W, X), flow (F) and procedure (C, A, D) has one identifier, listed in the Reading Companion's identifier register; the Test Strategy & Coverage document says which test proves each procedure.

How to read this document

Every account in the ledger is a **jar with a label**. Some jars hold money (at a partner, in NPL's wallet). Some hold promises (what NPL has undertaken to deliver, what someone owes NPL). Two hold NPL's results: what it has earned, and what it has cost. Every event moves value **out of one jar and into another**, and the amounts always match, so nothing ever appears from nowhere or disappears.

Each example is one event, shown as a table. The columns:

- **Account** — the jar, written as owner · place · currency · purpose. `CL` jars hold the client's money; `CO` jars are NPL's.
- **Debit / Credit** — the bookkeeper's two columns. You can ignore them.
- **In / Out** — the same thing in a word: IN means value goes *to* this jar, OUT means it comes *from* this jar. The word after the dot says what kind of jar it is.
- **Owner** — whose value it is, the client's or NPL's. **Component** — what kind of value: principal (the money itself), earnings, a cost, a fee, and so on.
- **Why** — the reason for the line, in a few words.

Read the **explanation on the right** first, then look for each sentence in the table. A tall table has its explanation above it instead.

THE JARS THAT APPEAR MOST

COLLECTED	client money that has arrived at a partner and is not yet converted
DUE	client money, converted, waiting at a partner to be paid out
HELD	client money sitting in one of NPL's own wallets (allowed in three cases only)
PAYABLE	the promise: what NPL still has to deliver for the client, per currency. Money already sent is out of NPL's hands and no longer here: "sent, not yet confirmed" is a mark on the payout line, not a jar
CREDIT / SHORTFALL	smaller promises: owed back to a sender; still owed to a receiver after a short delivery
POOL	NPL's own money resting at a partner, mostly its fees, invisible to the partner
WALLET	NPL's own money in its own wallets
EARN_GROSS / EARN_REBATE	what NPL has earned: its fees; rebates from partners, kept separate
EXP_...	what NPL has borne: partner fees, bank fees, network fees, rounding, losses
VAR_...	differences between what was expected and what happened, by cause
..._RECEIVABLE / ..._PAYABLE	what others owe NPL, and what NPL owes others, for shares, rebates, losses, invoices
FX_RESULT	NPL's own exchanges of one currency for another: what it sold and what it bought, per currency

Pattern A — Direct collection, conversion and accumulation

The everyday deal. A sender pays a partner, the partner converts the money, and the converted money waits at the partner until it is paid out. The fee is taken at the moment of conversion, and that is the only moment when some of the client's money becomes NPL's.

T1 · Collection of deal 1: 10,000 USDT arrive at Ali's USDT wallet endpoint (COLLECTION_RECEIVING_ENDPOINT, kind wallet) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	client money now sits at Ali, unconverted; the account is per partner, the endpoint only says which vehicle's wallet
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	NPL owes this onward on the client's behalf

WHAT IS HAPPENING

A sender has just paid 10,000 USDT into Ali's wallet. Two things are now true at once: the money is sitting at Ali (money IN to the "Ali, USDT" jar) and NPL has promised to deliver it onward for the client (a promise OUT of the "we owe the receivers" jar). Nothing is earned yet. The money is the client's; NPL is only holding it.

WHAT IS HAPPENING

Ali turns the USDT into euros. The USDT promise is torn up (IN · promise) and the USDT leaves Ali's USDT jar. What arrives in euros is split two ways: 8,820.00 into the client's EUR jar at Ali, and 90.00 EUR into NPL's own jar at Ali (its pool). A new promise is made in euros, 8,820.00, which is what the client was quoted. The 90.00 is NPL's fee: this is the one moment client money becomes NPL's. Ali's own cost is hidden inside its rate, so it never appears as a line.

THE ARITHMETIC

- 10,000.00 USDT × market 0.9000 = 9,000.00 EUR at market
- fee 2.00 % of that = 180.00 (sender pays 1.40 % = 126.00, receiver pays 0.60 % = 54.00)
- client net = 9,000.00 − 180.00 = 8,820.00 EUR → the client's jar and promise
- Ali converts at its own rate 0.891: 10,000.00 × 0.891 = 8,910.00 EUR delivered (its 1.00 % cost is inside the rate, never a line)
- NPL's margin = delivered 8,910.00 − client net 8,820.00 = 90.00 → NPL's pool at Ali

T2 · Conversion of deal 1 at Ali (fee_in_rate): 10,000 USDT → 8,910.00 EUR at 0.891 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		**IN** · promise	USDT	client	principal	the USDT obligation is discharged by the conversion
CL.ALI.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	the USDT left the collected position
CL.ALI.EUR.DUE	8,820.00		**IN** · money	EUR	client	principal	client net, waiting to be settled (EUR pairs round exact: cents are paid out)
CO.ALI.EUR.POOL	90.00		**IN** · money	EUR	company	earnings	NPL's margin, physically still at Ali (EARNINGS_RECEIVABLE)
CL.PROJECT.EUR.PAYABLE		8,820.00	**OUT** · promise	EUR	client	principal	owed to the receiver group, now in EUR
CO.NONE.EUR.EARN_GROSS		90.00	**OUT** · earned	EUR	company	earnings	margin recognised: the client's claim shrinks from 10,000 USDT to its EUR entitlement, the difference is NPL's

T3a · Same-currency collection at Ali: 5,000 EUR by bank transfer (as T1) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.EUR.COLLECTED	5,000.00		**IN** · money	EUR	client	principal	
CL.PROJECT.EUR.PAYABLE		5,000.00	**OUT** · promise	EUR	client	principal	

WHAT IS HAPPENING

A plain bank transfer of 5,000 EUR arrives at Ali for a deal where the sender pays in euros and the receiver gets euros. Same shape as T1: money IN at Ali, a promise OUT.

WHAT IS HAPPENING

Even though no currency changes, the deal still "converts" at a rate of 1. That is how the fee is taken. Evo's EUR → EUR fee is 2.8 %, split half and half (sender 1.4 %, receiver 1.4 %): of the 5,000, 4,860.00 is the client's (promise and jar in euros) and 140.00 is NPL's fee into its pool. Without this step there would be no moment at which the fee is earned.

THE ARITHMETIC

- 5,000 EUR × rate 1 = 5,000 at market
- EUR → EUR fee 2.80 % = 140.00 (sender 1.40 % = 70.00, receiver 1.40 % = 70.00)
- client net = 4,860.00 → client's jar and promise; NPL's margin 140.00 → pool

T3 · Same-currency pass-through: the 5,000 EUR converted at rate 1 — Evo's EUR → EUR fee 2.80 % (sender pays 1.40 %, receiver pays 1.40 %) captured, client net 4,860.00 EUR (the collection posted as in T1 and is not repeated) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.EUR.PAYABLE	5,000.00		**IN** · promise	EUR	client	principal	the obligation is re-stated at the sender's price
CL.ALI.EUR.COLLECTED		5,000.00	**OUT** · money	EUR	client	principal	
CL.ALI.EUR.DUE	4,860.00		**IN** · money	EUR	client	principal	client net, available to settle
CO.ALI.EUR.POOL	140.00		**IN** · money	EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,860.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		140.00	**OUT** · earned	EUR	company	earnings	the fee is earned at conversion even though no currency changes

Pattern B — Part collection

A sender may pay a deal in several instalments. Each instalment is recorded as it arrives; the conversion waits until the deal is fully paid.

T4 · Collection of deal 2, part 1 of 2: 4,000 USDT

source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	4,000.00		**IN** · money	USDT	client	principal	first part
CL.PROJECT.USDT.PAYABLE		4,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

The sender pays a deal of 5,000 USDT in two instalments. The first 4,000 arrive: money IN at Ali, promise OUT for the same amount. The deal is not complete, so nothing converts yet.

T5 · Collection of deal 2, part 2 of 2: 1,000 USDT (next day)

source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	1,000.00		**IN** · money	USDT	client	principal	second part completes the deal
CL.PROJECT.USDT.PAYABLE		1,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

The remaining 1,000 arrive the next day. Same two lines. Now the whole 5,000 is at Ali and the promise is 5,000.

WHAT IS HAPPENING

With the deal fully paid, Ali converts the whole 5,000 at once, exactly like T2: USDT out, euros in for the client, NPL's fee into its pool, the promise re-stated in euros. A cash deal would wait here until the partner has counted the notes.

THE ARITHMETIC

- 5,000.00 USDT × market 0.9000 = 4,500.00 EUR at market
- fee 2.00 % of that = 90.00 (sender pays 1.40 % = 63.00, receiver pays 0.60 % = 27.00)
- client net = 4,500.00 − 90.00 = 4,410.00 EUR → the client's jar and promise
- Ali converts at its own rate 0.891: 5,000.00 × 0.891 = 4,455.00 EUR delivered (its 1.00 % cost is inside the rate, never a line)
- NPL's margin = delivered 4,455.00 − client net 4,410.00 = 45.00 → NPL's pool at Ali

T6 · Conversion of deal 2 at Ali: 5,000 USDT → 4,455.00 EUR source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	5,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		5,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	4,410.00		**IN** · money	EUR	client	principal	
CO.ALI.EUR.POOL	45.00		**IN** · money	EUR	company	earnings	
CL.PROJECT.EUR.PAYABLE		4,410.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		45.00	**OUT** · earned	EUR	company	earnings	

Pattern C — Settlement, confirmation, bank fees, shortfalls, rounding

Paying out: the partner sends the money, and from that moment it is out of NPL's hands, so the promise is treated as kept. The receiver's confirmation changes nothing unless it is short; a short delivery or a return brings the promise back for the missing part. Bank fees and rounding differences are settled according to the rules of the project. Bank transfers go out exact, cents included; rounding to a unit applies to cash hand-overs.

T7 · Settlement from Ali: 18,090.00 EUR released in two lines (Entity X1 8,000.00, Entity X2 10,090.00); the promise is discharged on release source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.EUR.PAYABLE	18,090.00		**IN** · promise	EUR	client	principal	once the money has left Ali it is no longer in NPL's hands, so NPL no longer owes it; a return or a short delivery re-raises the promise
CL.ALI.EUR.DUE		18,090.00	**OUT** · money	EUR	client	principal	both lines leave the client's balance at Ali; EUR bank transfers carry the cents (amount_rounding exact)

WHAT IS HAPPENING

Ali now pays out. The client's euro jar at Ali holds several converted deals, 18,090.00 in all; NPL sends it in two bank transfers, 8,000.00 to one of the receiver's accounts and 10,090.00 to another, cents included. The moment the money leaves Ali it is out of NPL's hands, so the promise is discharged right away: money OUT of the Ali jar, promise IN (fulfilled). The two transfers are now "sent"; the system remembers that on the payout lines, not in a jar. If one comes back, the promise comes back with it (T24).

THE ARITHMETIC

- client euros at Ali: deal 1 8,820.00 + deal 2 4,410.00 + pass-through 4,860.00 = 18,090.00
- paid out in full, cents included: 8,000.00 + 10,090.00 = 18,090.00; promise discharged by the same 18,090.00

T8 · Confirmation, line 1 full: Entity X1 received 8,000.00 EUR source: Confirmation

WHAT IS HAPPENING

The receiver confirms the first transfer arrived in full. Nothing posts: the promise was already discharged when the money was sent. The payout line just changes from "sent" to "confirmed", and a control checks that every sent line ends up confirmed, short or returned.

T9 · Confirmation, line 2 short: Entity X2 received 10,040.00 EUR, the bank took 50.00 (treatment carry_forward) source: Confirmation

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.RECEIVER_X.EUR.SHORTFALL	50.00		**IN** · money	EUR	client	bank_fee	still owed to the receiver, topped up at the next settlement
CL.PROJECT.EUR.PAYABLE		50.00	**OUT** · promise	EUR	client	bank_fee	the promise is re-raised for the part that did not arrive

WHAT IS HAPPENING

The second transfer arrives 50.00 short because a bank took a fee on the way. The 10,040.00 that arrived needs no posting. For the missing 50.00 the promise is re-raised (NPL owes it again) and the amount is parked in a "shortfall" jar for that receiver, to be topped up next time. If the client had agreed to bear bank fees, nothing would post at all.

THE ARITHMETIC

- line 2 was 10,090.00; received 10,040.00; bank fee 50.00 → shortfall jar, promise re-raised by 50.00

T10 · Shortfall top-up funded from NPL's pool at Ali (BANK_FEE_EVENT applied to the next settlement) source: BankFeeEvent

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.EUR.EXP_BANKFEE	50.00		**IN** · cost	EUR	company	bank_fee	NPL bears the cost
CO.ALI.EUR.POOL		50.00	**OUT** · money	EUR	company	bank_fee	paid out of NPL's money resting at Ali
CL.ALI.EUR.DUE	50.00		**IN** · money	EUR	client	bank_fee	the 50 EUR is client money again, in balance at Ali
CL.RECEIVER_X.EUR.SHORTFALL		50.00	**OUT** · money	EUR	client	bank_fee	the shortfall is cleared

WHAT IS HAPPENING

NPL makes the shortfall good from its own money at Ali. 50.00 goes IN to NPL's cost jar (a bank fee NPL bears) and OUT of NPL's pool at Ali; the same 50.00 goes back IN to the client's jar at Ali and the shortfall jar is emptied. This is the rare direction: NPL's money becoming client money. It goes out with the next payout like any other balance.

THE ARITHMETIC

- 50.00 from NPL's pool at Ali → NPL's cost jar; the same 50.00 into the client's jar, shortfall cleared

WHAT IS HAPPENING

A Thai-baht payout handed over in cash, which is rounded up to the nearest 1,000 baht (a bank transfer in baht would go out exact). The client is owed 95,844.00 but will be paid 96,000.00. The extra 156.00 is NPL's cost (rounding jar) and comes out of NPL's margin, so NPL's pool gets 810.00 instead of 966.00. Rounding down (a truncating option on a cash pair) is the opposite case: the dropped fraction stays the client's; T2 is a EUR bank transfer and goes out exact.

THE ARITHMETIC

- 3,000 USDT × market 32.60 = 97,800.00 THB; fee 2.00 % = 1,956.00; client net 95,844.00
- Ali converts at 32.27: 96,810.00 delivered; margin 966.00
- round up to the next 1,000: payout 96,000.00; the extra 156.00 comes out of NPL's margin → rounding cost; pool gets 810.00

T11 · Conversion for a cash payout with round-up: 3,000 USDT → 96,810.00 THB at 32.27 (market 32.6); client net 95,844.00, handed over in cash as 96,000.00 (amount_rounding round_up_n, rounding_unit 1,000 — the payout option; cash_rounding is the collection side) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	(collection of the 3,000 USDT assumed posted as T1)
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.THB.DUE	95,844.00		**IN** · money	THB	client	principal	client net as priced
CL.ALI.THB.DUE	156.00		**IN** · money	THB	client	rounding	the amount added by rounding up: client money from now on
CO.ALI.THB.POOL	810.00		**IN** · money	THB	company	earnings	NPL's margin less the round-up it funded
CO.NONE.THB.EXP_ROUNDING	156.00		**IN** · cost	THB	company	rounding	the cost of rounding up (rounding_residual is negative)
CL.PROJECT.THB.PAYABLE		96,000.00	**OUT** · promise	THB	client	principal	the obligation is the rounded payout
CO.NONE.THB.EARN_GROSS		966.00	**OUT** · earned	THB	company	earnings	margin as priced

Pattern D — Contingency routes and the bridge wallet

Sometimes money has to pass through one of NPL's own wallets: when the usual partner is unavailable, or when the sender can only pay on a network the partner does not accept. The ledger tracks it there until it leaves.

T12 · Deal 3, Ali unavailable: 10,000 USDT collected into NPL's contingency wallet under an approved REROUTE

source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWN.USDT.HELD	10,000.00		**IN** · money	USDT	client	principal	client money in custody, with deal, reason and age
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

Ali is unavailable, so with management's approval the sender pays 10,000 USDT into NPL's own contingency wallet instead. The money goes IN to a "held in NPL's wallet" jar, and the usual promise goes OUT. Money in NPL's own wallet is allowed only in three named cases; this is one of them, and a custody report watches it until it leaves.

T13 · Forward to Jeton with recover dues: 10,000 USDT of client money credited at Jeton, 30.00 USDT of NPL's dues netted

source: Reroute

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.JETON.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	Jeton credits the full client amount
CL.OWN.USDT.HELD		10,000.00	**OUT** · money	USDT	client	principal	custody ends
CO.OWN.USDT.WALLET	30.00		**IN** · money	USDT	company	earnings	NPL keeps 30 USDT in its wallet instead of sending it
CO.JETON.USDT.POOL		30.00	**OUT** · money	USDT	company	earnings	Jeton's debt to NPL falls by the same 30 USDT

WHAT IS HAPPENING

NPL forwards the client's money to Jeton. The held jar empties and the "at Jeton, USDT" jar fills with the full 10,000. But NPL physically sends only 9,970.00: Jeton already owed NPL 30.00 of earlier fees, so they net it off. For NPL that is 30.00 IN to its own wallet (kept instead of sent) and 30.00 OUT of what Jeton owed it. Nothing is taken from the client's amount.

THE ARITHMETIC

- client amount credited at Jeton 10,000; NPL sends 9,970.00 and keeps 30.00, which Jeton owed NPL from earlier deals

WHAT IS HAPPENING

Jeton converts at the market rate and then charges its own fee separately (1.25 %). So the lines show three things: the client's euros (8,820.00) into the client's jar at Jeton, Jeton's fee (112.50) into NPL's cost jar, and NPL's net margin (67.50) into its pool at Jeton. The fee NPL charged in total was 180.00 (the "earned" line); Jeton's cost comes out of it. Compare T2, where Ali's cost is invisible because it is inside Ali's rate.

THE ARITHMETIC

- 10,000.00 USDT × market 0.9000 = 9,000.00 EUR at market
- fee 2.00 % of that = 180.00 (sender pays 1.40 % = 126.00, receiver pays 0.60 % = 54.00)
- client net = 9,000.00 − 180.00 = 8,820.00 EUR → the client's jar and promise
- Jeton converts at market: 9,000.00, then charges 1.25 % = 112.50 → NPL's cost jar; delivered 8,887.50
- NPL's margin = delivered 8,887.50 − client net 8,820.00 = 67.50 → NPL's pool at Jeton; fee charged 180.00 − partner cost 112.50 = 67.50

T14 · Conversion of deal 3 at Jeton (fee_on_market, 1.25 %): 10,000 USDT → 9,000.00 EUR at market, Jeton keeps 112.50 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		**IN** · promise	USDT	client	principal	
CL.JETON.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	
CL.JETON.EUR.DUE	8,820.00		**IN** · money	EUR	client	principal	client net
CO.JETON.EUR.POOL	67.50		**IN** · money	EUR	company	earnings	NPL's net margin resting at Jeton
CO.NONE.EUR.EXP_PARTNER	112.50		**IN** · cost	EUR	company	partner_cost	Jeton's stated fee, deducted at transaction level
CL.PROJECT.EUR.PAYABLE		8,820.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		180.00	**OUT** · earned	EUR	company	earnings	gross margin before partner cost

T15 · Deal 4: the sender can only pay on ERC-20, which Ali does not accept; 10,000 USDT arrive on NPL's bridge wallet source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWN.USDT.HELD	10,000.00		**IN** · money	USDT	client	principal	client money on the bridge (OWN_WALLET.role collection_bridge), to be forwarded within custody_max_hours
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

This sender can only pay on a network Ali does not accept. So NPL lets the money land on its own "bridge" wallet first: money IN to the held jar, promise OUT. It must be forwarded within a set number of hours.

T16 · Hop sent: the bridge forwards 10,000 USDT to Ali on TRC-20 (a SETTLEMENT_LINE to the partner_transit receiver, outcome sent) source: Settlement

> **WHAT IS HAPPENING**
>
> The bridge forwards the 10,000 to Ali on a network Ali does accept. Nothing posts yet: a hop is not a delivery, so the promise stays, and the money stays in the held jar until Ali says it has arrived. The payout line is marked "sent".

T17 · Ali credits the hop: 10,000 USDT now sit at Ali as collected client money; network fee 4.20 USDT paid by NPL when one applies source: Confirmation

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	the leg continues at its partner exactly as T1
CL.OWN.USDT.HELD		10,000.00	**OUT** · money	USDT	client	principal	custody ends
CO.NONE.USDT.EXP_NETWORK	4.20		**IN** · cost	USDT	company	network_fee	network fee of the hop, in USDT, NPL's cost (project network_fee_policy)
CO.OWN.USDT.WALLET		4.20	**OUT** · money	USDT	company	network_fee	paid from NPL's own USDT; when no network fee applies these two lines are absent

> **WHAT IS HAPPENING**
>
> Ali confirms the money has arrived. The held jar empties and the client's USDT jar at Ali fills, which is exactly where T1 would have put it. If the network charged a fee for the transfer, NPL pays it in USDT from its own wallet: IN to the cost jar, OUT of NPL's wallet; if no fee applies, those two lines simply do not exist. Either way the client pays nothing for the hop, and from here the deal continues normally.

Pattern E — Cancellations, reversals and voids

Two different things happen here. **A mistake in the books** (T18, T19): a posting was wrong, say a collection booked against the wrong deal. The ledger never lets anyone edit or delete it, because then yesterday's reports would no longer add up. Instead a second posting is made that is the exact mirror of the first and points at it, so the two cancel, and then the correct posting is made. Three entries stay on record: the wrong one, its cancellation, the right one. **A cancelled deal** (T20, T21): the money really did arrive, but the deal is called off, so the money has to go back to the sender. The promise to deliver is torn up and the money leaves the partner. Sending crypto costs a network fee; the project's rules say whether the sender bears it (the refund is a little smaller) or NPL does (the refund is whole and NPL books the fee as its cost). Because money has already moved, Management must approve how a cancelled deal is unwound.

T18 · Erroneous posting: a collection of 2,000 USDT was booked against the wrong deal (as posted) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	2,000.00		**IN** · money	USDT	client	principal	booked in error
CL.PROJECT.USDT.PAYABLE		2,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

A mistake: someone records a 2,000 USDT collection against the wrong deal. The posting looks normal (money IN at Ali, promise OUT), but it is wrong.

T19 · Reversal of T18 (LEDGER_TRANSACTION.reversal_of_id = T18), then T18 is re-posted correctly source: Adjustment

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	2,000.00		**IN** · promise	USDT	client	principal	mirror image of T18
CL.ALI.USDT.COLLECTED		2,000.00	**OUT** · money	USDT	client	principal	

WHAT IS HAPPENING

Nothing is ever deleted. Instead a mirror-image transaction is posted that points at the wrong one: promise IN, money OUT, same amounts. The two cancel. Then the correct posting is made afresh. Anyone looking at history sees the mistake, the reversal and the fix, so the books add up at every past moment.

T20 · Collection of deal 5: 3,000 USDT arrive at Ali (the deal is voided afterwards) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	3,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		3,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

A sender pays 3,000 USDT for a deal that is about to be cancelled. Normal collection lines.

T21 · Deal 5 voided after collection: refund to the sender in USDT, network fee 2.00 USDT charged back to the sender (project network_fee_policy) source: Adjustment

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	the obligation ends
CL.ALI.USDT.COLLECTED		2,998.00	**OUT** · money	USDT	client	principal	returned to the sender's wallet
CL.ALI.USDT.COLLECTED		2.00	**OUT** · money	USDT	client	network_fee	paid to the network; the sender bears it

WHAT IS HAPPENING

The deal is voided, so the money must go back. The promise is torn up (IN · promise, the obligation ends) and the money leaves Ali's jar: 2,998.00 back to the sender and 2.00 to the network as the sending fee, which this project charges to the sender. If NPL absorbed the fee instead, the sender would get the full 3,000 and the 2.00 would come out of NPL's pool into its cost jar.

THE ARITHMETIC

- refund 3,000.00: 2,998.00 to the sender, 2.00 network fee charged to the sender

Every other kind of mistake is fixed the same way as T19: a mirror-image posting that points at the wrong one, then the right posting, both dated today. A posting made twice, money sent to the wrong receiver, a conversion booked in the wrong currency, a correction of only part of an amount, a mistake found after the month has been reported to the accountant: each is a reversal and a re-posting, labelled with the month it belongs to when that month has already been reported. The worked examples will be added when the first real case happens during the parallel run.

Pattern F — Rejected payout and return leg

A bank can reject a payout. The money comes back, the failed payout is undone, and the client can either try again later or have the money converted back and returned.

T22a · Collection of deal 6 at Ali: 2,000 USDT (as T1) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	2,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		2,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

A deal of 2,000 USDT that will be paid out in pounds. Money IN at Ali, promise OUT.

WHAT IS HAPPENING

Ali converts to pounds, same shape as T2: USDT out, pounds in for the client (1,528.80), NPL's margin (15.60) into its pool, promise re-stated in pounds.

THE ARITHMETIC

- 2,000 USDT × market 0.7800 = 1,560.00 GBP; fee 2.00 % = 31.20; client net 1,528.80
- Ali converts at 0.7722: 1,544.40 delivered; margin 15.60 → pool

T22 · Conversion of deal 6 at Ali: 2,000 USDT → 1,544.40 GBP (the collection posted as in T1 and is not repeated) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	2,000.00		**IN** · promise	USDT	client	principal	USDT leg
CL.ALI.USDT.COLLECTED		2,000.00	**OUT** · money	USDT	client	principal	USDT leg
CL.ALI.GBP.DUE	1,528.80		**IN** · money	GBP	client	principal	
CO.ALI.GBP.POOL	15.60		**IN** · money	GBP	company	earnings	
CL.PROJECT.GBP.PAYABLE		1,528.80	**OUT** · promise	GBP	client	principal	
CO.NONE.GBP.EARN_GROSS		15.60	**OUT** · earned	GBP	company	earnings	

T23 · Settlement of deal 6: 1,528.80 GBP released to the receiver's bank (the line names deal 6 in deal_id); the promise is discharged on release source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.GBP.PAYABLE	1,528.80		**IN** · promise	GBP	client	principal	sent: NPL no longer owes it unless it comes back
CL.ALI.GBP.DUE		1,528.80	**OUT** · money	GBP	client	principal	

WHAT IS HAPPENING

The pounds are sent to the receiver's bank: money OUT of the client's jar at Ali and, because sent money is out of NPL's hands, the promise is discharged at the same time (IN · promise).

WHAT IS HAPPENING

The receiver's bank rejects the transfer and sends it back, minus a 15.00 fee. What came back goes into the client's jar at Ali and the promise is re-raised for it: NPL owes it again. The 15.00 the bank kept is NPL's cost; NPL puts that much from its own pool into the client's jar and re-raises the promise for it too, so the client is exactly where it was before the payout. If NPL's pool at Ali holds no pounds, the posting is the same and the pool simply goes below zero for a while: Ali fronted the 15.00 out of the one balance it holds. T24a shows how that is squared.

THE ARITHMETIC

- sent 1,528.80; came back 1,513.80 → client's jar, promise re-raised by the same
- bank kept 15.00 → NPL's cost; NPL puts 15.00 from its pool into the client's jar and re-raises the promise, so the client has 1,528.80 owed and in its jar again

T24 · The receiving bank rejects the transfer: 1,513.80 GBP come back to Ali, the bank kept 15.00 (SETTLEMENT_RETURN; line outcome = returned) source: SettlementReturn

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.GBP.DUE	1,513.80		**IN** · money	GBP	client	principal	back in the client balance at Ali
CL.PROJECT.GBP.PAYABLE		1,513.80	**OUT** · promise	GBP	client	principal	the promise is re-raised: NPL owes it again
CO.NONE.GBP.EXP_BANKFEE	15.00		**IN** · cost	GBP	company	bank_fee	bounce fee absorbed by NPL (treatment absorbed)
CO.ALI.GBP.POOL		15.00	**OUT** · money	GBP	company	bank_fee	funded from NPL's pool at Ali; if the GBP pool cannot cover it, it goes negative until T24a
CL.ALI.GBP.DUE	15.00		**IN** · money	GBP	client	bank_fee	NPL makes the client whole
CL.PROJECT.GBP.PAYABLE		15.00	**OUT** · promise	GBP	client	bank_fee	and the promise is whole again

T24a · Squaring NPL's GBP pool at Ali: with the client's next USDT conversion Ali also converts 19.43 USDT of NPL's own (its USDT pool there) into 15.00 GBP at 0.7722 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.USDT.FX_RESULT	19.43		**IN** · cost	USDT	company	earnings	currency sold: NPL's USDT leave its pool at Ali
CO.ALI.USDT.POOL		19.43	**OUT** · money	USDT	company	earnings	NPL's earnings kept in USDT at Ali (Pattern N) fund it
CO.ALI.GBP.POOL	15.00		**IN** · money	GBP	company	earnings	NPL's GBP pool is back to where it was before the bounce
CO.NONE.GBP.FX_RESULT		15.00	**OUT** · cost	GBP	company	earnings	currency bought: the GBP NPL received for them

WHAT IS HAPPENING

NPL rarely has a pool in every currency at every partner; its earnings mostly sit as USDT (Pattern N). So when it has absorbed a fee in pounds and its pound pool is below zero, it squares the pool at the next conversion: Ali converts a little of NPL's own USDT along with the client's money. 19.43 USDT of NPL's leave its USDT pool at Ali and 15.00 GBP arrive in its GBP pool. The two "currency result" lines record that NPL sold USDT and bought pounds; that jar is NPL's own exchange record and needs nothing done to it later. The partner sees one ordinary conversion. The same mechanism pays for the 50 EUR top-up in T10 if NPL had no euros at Ali.

THE ARITHMETIC

- 15.00 GBP ÷ Ali's rate 0.7722 = 19.43 USDT of NPL's own converted with the next client conversion
- NPL's USDT pool at Ali −19.43; NPL's GBP pool at Ali +15.00; FX_RESULT records USDT sold and GBP bought

T25 · Return leg: 1,528.80 GBP converted back to 1,964.51 USDT at Ali (rate 1.285); markup waived by FEE_DECISION source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.GBP.PAYABLE	1,528.80		**IN** · promise	GBP	client	principal	GBP obligation ends
CL.ALI.GBP.DUE		1,528.80	**OUT** · money	GBP	client	principal	
CL.ALI.USDT.DUE	1,964.51		**IN** · money	USDT	client	principal	USDT owed to the sender's return receiver
CL.PROJECT.USDT.PAYABLE		1,964.51	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

The client decides to convert the pounds back to USDT and have them returned. The pound promise is torn up and the pounds leave Ali; 1,964.51 USDT arrive in the client's jar with a new USDT promise. NPL waived its fee on this return, so there is no earned line. Note NPL's margin from T22 is untouched: it was earned at the first conversion and a return never claws it back.

THE ARITHMETIC

- 1,528.80 GBP × 1.2850 = 1,964.51 USDT; no fee (waived), so the whole amount is the client's

Pattern H — Balance conversion

Money already converted and waiting at a partner can be converted again into another currency, without a new sender or receiver, when a payout needs it.

T26 · Opening position: 5,000 SGD of converted client balance at Ali from earlier deals (summarised)

source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.SGD.DUE	5,000.00		**IN** · money	SGD	client	principal	balance built up by earlier SGD deals
CL.PROJECT.SGD.PAYABLE		5,000.00	**OUT** · promise	SGD	client	principal	

WHAT IS HAPPENING

Starting point for the next example: the client has 5,000 Singapore dollars sitting at Ali from earlier deals (money in the SGD jar, promise in SGD).

T27 · Balance conversion at Ali: 5,000 SGD of client balance → 3,700.00 USDT at 0.74, no markup (serves Group B)

source: BalanceConversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.SGD.PAYABLE	5,000.00		**IN** · promise	SGD	client	principal	SGD obligation ends
CL.ALI.SGD.DUE		5,000.00	**OUT** · money	SGD	client	principal	SGD balance leaves
CL.ALI.USDT.DUE	3,700.00		**IN** · money	USDT	client	principal	USDT balance arrives
CL.PROJECT.USDT.PAYABLE		3,700.00	**OUT** · promise	USDT	client	principal	same obligation, new currency

WHAT IS HAPPENING

NPL has the balance itself converted to USDT, without any sender or receiver involved, because a receiver needs USDT. The SGD promise is torn up and the SGD leave; 3,700.00 USDT arrive in the client's jar with a USDT promise. No fee here (that would be a separate decision). Finance asks for this, Management approves it. Like a deal leg, a banking-ready balance waits for the partner's cutoff rules before it converts.

THE ARITHMETIC

- 5,000 SGD × 0.7400 = 3,700.00 USDT; no fee; USDT pairs round exact

Pattern I — Overpayment and sender credit

A sender who pays too much gets a credit, which is refunded or used on the next deal.

T28 · Overpayment: a sender sends 10,050 USDT against a 10,000 USDT deal; 50 USDT becomes sender credit

source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,050.00		**IN** · money	USDT	client	principal	all of it is client money at Ali
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	the deal
CL.SENDER_A.USDT.CREDIT		50.00	**OUT** · promise	USDT	client	principal	owed back to the sender (SENDER_CREDIT), or applied to the next deal

WHAT IS HAPPENING

A sender pays 10,050 USDT against a 10,000 deal. All 10,050 are client money at Ali. The promise for the deal is 10,000; the extra 50 becomes a separate promise back to the sender, a credit.

THE ARITHMETIC

- received 10,050; deal 10,000; credit to the sender 50

T29 · The credit is applied to the sender's next deal of 5,000 USDT: the sender pays 4,950

source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	4,950.00		**IN** · money	USDT	client	principal	what actually arrives
CL.SENDER_A.USDT.CREDIT	50.00		**IN** · promise	USDT	client	principal	the credit is consumed (state applied)
CL.PROJECT.USDT.PAYABLE		5,000.00	**OUT** · promise	USDT	client	principal	the full deal is collected

WHAT IS HAPPENING

The sender's next deal is 5,000 but they only need to send 4,950: the 50 credit is used up. The credit jar empties (IN · promise, the promise to the sender is fulfilled) and together with the 4,950 that arrived, the new deal's promise is the full 5,000.

THE ARITHMETIC

- new deal 5,000 = 4,950 received + 50 credit used

Pattern J — Markup shares and partner rebates

Two kinds of side payments, recorded the moment the conversion happens and settled monthly against a statement. A share is part of NPL's fee owed to a markup-share party — a customer such as Raeen, or an outside introducer; a rebate is money a partner gives back to NPL, kept separate and never shared.

T30 · Markup share, npl_pays: an external markup-share party earns 25 % of NPL's earnings on deal 3 = 16.88 EUR, accrued with the conversion

source: MarkupShare

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.EUR.EARN_GROSS	16.88		**IN** · earned	EUR	company	share	reduces NPL's margin
CO.PARTY_P.EUR.SHARE_PAYABLE		16.88	**OUT** · promise	EUR	company	share	owed to the party (MARKUP_SHARE_ACCRUAL, direction payable)

WHAT IS HAPPENING

An outside party who brought in this business is owed a quarter of NPL's margin on deal 3. The share, 16.88, is taken off NPL's earned jar (OUT · earned, the margin is smaller) and parked as a promise to that party. It is booked at the same moment as the conversion.

THE ARITHMETIC

- NPL's margin on deal 3 (T14) 67.50 × 25 % = 16.88 → owed to the party, off NPL's earned jar

T31 · Month-end: NPL pays the party 16.88 EUR from its own wallet

source: MarkupShare

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.PARTY_P.EUR.SHARE_PAYABLE	16.88		**IN** · promise	EUR	company	share	accrual state settled, paid_at set
CO.OWN.EUR.WALLET		16.88	**OUT** · money	EUR	company	share	or from the pool at a partner, when the partner pays the party on NPL's instruction

WHAT IS HAPPENING

At month end NPL pays the party: the promise is fulfilled (IN · promise) and the money leaves NPL's own wallet.

WHAT IS HAPPENING

A customer called Raeen brings deals through its own sub-account at Aquanow and shares NPL's margin half and half. The sender paid a 4 % fee; Aquanow charges 0.80 %; what is left to share is 288.00. The client's euros go into the client's jar as usual. NPL's half, 144.00, is not sitting in a pool at Aquanow because Raeen holds the converted money, so instead it is a promise Raeen owes NPL (a receivable). Raeen's half, 144.00, is taken off NPL's earned jar. Raeen never learns about the rebate in T33. Which side holds NPL's share is recorded, not assumed: the share rule says whether NPL pays the party or the party holds the money and pays NPL, and every accrual is tagged payable or receivable per deal, so the same party can be on either side depending on the route.

THE ARITHMETIC

- 10,000 USDT × market 0.9000 = 9,000.00; sender pays 4 % = 360.00; client net 8,640.00
- Aquanow converts at market and charges 0.80 % = 72.00; delivered 8,928.00
- share base = fee 360.00 − partner cost 72.00 = 288.00; Raeen 50 % = 144.00; NPL 50 % = 144.00 (owed by Raeen, who holds the money)

T32 · Raeen sub-account at Aquanow (party_retains): 10,000 USDT → 8,928.00 EUR; sender pays 4 % on market, Aquanow charges 0.80 %; net markup 288.00 EUR shared 50 / 50 source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	10,000.00		**IN** · promise	USDT	client	principal	
CL.AQN.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	
CL.AQN.EUR.DUE	8,640.00		**IN** · money	EUR	client	principal	client net, in Raeen's sub-account at Aquanow, settled to Raeen's receivers as usual
CO.RAEEN.EUR.SHARE_RECEIVABLE	144.00		**IN** · money	EUR	company	share	NPL's remainder: Raeen holds the converted funds and owes it (direction receivable, period)
CO.NONE.EUR.EARN_GROSS	144.00		**IN** · earned	EUR	company	share	Raeen's share of the net markup; never NPL's money
CO.NONE.EUR.EXP_PARTNER	72.00		**IN** · cost	EUR	company	partner_cost	Aquanow's stated fee
CL.PROJECT.EUR.PAYABLE		8,640.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		360.00	**OUT** · earned	EUR	company	earnings	gross margin: the sender's 4 %

T33 · Partner rebate accrued with the same conversion: Aquanow returns 0.3 % of the amount = 30.00 USDT source: Rebate

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.AQN.USDT.REBATE_RECEIVABLE	30.00		**IN** · money	USDT	company	rebate	expected from Aquanow (PARTNER_REBATE_ACCRUAL state expected, period)
CO.NONE.USDT.EARN_REBATE		30.00	**OUT** · earned	USDT	company	rebate	NPL's alone: never in EARN_GROSS, never visible to Raeen

WHAT IS HAPPENING

Aquanow gives NPL back 0.3 % of the amount as a rebate: 30.00 USDT. NPL records it straight away as something Aquanow owes (a receivable) and as income in a separate "rebate earned" jar, never mixed with the margin, and never part of what is shared with Raeen.

THE ARITHMETIC

- rebate 0.3 % × 10,000 USDT = 30.00 USDT, NPL's alone

T34 · Month-end: Aquanow's statement shows 29.50 USDT for the deal and pays it in USDT; the 0.50 USDT difference trues up the income

source: Rebate

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.OWN.USDT.WALLET	29.50		**IN** · money	USDT	company	rebate	received (statement_ref on the accrual, state received)
CO.NONE.USDT.EARN_REBATE	0.50		**IN** · earned	USDT	company	rebate	true-up against the statement: the accrual's received_amount
CO.AQN.USDT.REBATE_RECEIVABLE		30.00	**OUT** · money	USDT	company	rebate	the expectation is cleared

WHAT IS HAPPENING

Aquanow's monthly statement says 29.50, a little less than expected. NPL receives 29.50 into its wallet, clears the whole 30.00 it had expected, and the 0.50 difference is simply written off the rebate income. Nothing from before is edited.

THE ARITHMETIC

- expected 30.00; statement 29.50; difference 0.50 written off the rebate income

T35 · Month-end: Raeen pays NPL its remainder 144.00 EUR

source: MarkupShare

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.OWN.EUR.WALLET	144.00		**IN** · money	EUR	company	share	received
CO.RAEEN.EUR.SHARE_RECEIVABLE		144.00	**OUT** · money	EUR	company	share	accrual state settled, received_amount

WHAT IS HAPPENING

Raeen pays NPL its half: money IN to NPL's wallet, the receivable cleared.

Pattern K — Variances, by kind

The client's figures are fixed when the sender is quoted. Whenever the partner's actual conversion differs from the one the quote assumed, the difference is NPL's, and it is filed in one of four jars according to why it happened.

WHAT IS HAPPENING

Ali converted at a slightly worse rate than the one it had quoted. The client is not affected: their euros were fixed when the sender was quoted. The difference (1.50) is NPL's and goes into a variance jar labelled "the partner did something different from expected". NPL's pool at Ali gets that much less.

THE ARITHMETIC

- 3,000 USDT; client net at market 2,646.00 (fixed at quote)
- expected from Ali at 0.8910: 2,673.00; Ali actually booked at 0.8905: 2,671.50
- margin as quoted 27.00; variance 1.50 → "partner did something different" jar; pool gets 25.50

T36 · Partner calculates (calc_mode partner_calculates): Ali booked 3,000 USDT at 0.8905 instead of the 0.8910 version: expected 2,673.00 EUR, actual 2,671.50 EUR, variance 1.50 EUR source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	2,646.00		**IN** · money	EUR	client	principal	client net is unchanged: the sender's price is write-once
CO.ALI.EUR.POOL	25.50		**IN** · money	EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_CONVERSION	1.50		**IN** · cost	EUR	company	variance	what the partner did worse than expected; beyond the project's tolerance an exception opens, the deal does not wait
CL.PROJECT.EUR.PAYABLE		2,646.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		27.00	**OUT** · earned	EUR	company	earnings	margin as priced

WHAT IS HAPPENING

The rate was locked after Ali's 14:30 cutoff, so the conversion ran the next business day at a different rate. Same shape as T36, but the difference goes into a separate jar labelled "cutoff timing", because NPL chose to absorb this kind of difference and wants to see it on its own. NPL does not have to accept a bad next-day rate: it can push the conversion a further day with a stated reason, or ask Ali for a better rate; nothing posts until the conversion actually runs.

THE ARITHMETIC

- expected at the locked 0.8910: 2,673.00; converted next day at 0.8890: 2,667.00; difference 6.00 → cutoff-timing jar

T37 · Cutoff timing: 3,000 USDT locked at 0.8910 after Ali's 14:30 cutoff, converted next business day at 0.8890: expected 2,673.00 EUR, actual 2,667.00 EUR, variance 6.00 EUR source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	2,646.00		**IN** · money	EUR	client	principal	client net is unchanged: the sender's price is write-once
CO.ALI.EUR.POOL	21.00		**IN** · money	EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_CUTOFF	6.00		**IN** · cost	EUR	company	cutoff_timing	rate stamped at lock vs rate used; NPL absorbs it
CL.PROJECT.EUR.PAYABLE		2,646.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		27.00	**OUT** · earned	EUR	company	earnings	margin as priced

WHAT IS HAPPENING

A sender's quote had expired, and NPL agreed to honour yesterday's rate anyway. Ali converts at today's rate. The difference is NPL's cost by choice and goes into its own jar, "honoured rate", so management can see what honouring old quotes costs.

THE ARITHMETIC

- honoured rate 0.8910: 2,673.00; today's rate 0.8860: 2,658.00; difference 15.00 → honoured-rate jar

T38 · Honoured expired rate: a new leg priced on yesterday's 0.8910 by an approved override (honour_expired_rate); Ali converts today at 0.8860: expected 2,673.00 EUR, actual 2,658.00 EUR, variance 15.00 EUR source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,000.00		**IN** · promise	USDT	client	principal	
CL.ALI.USDT.COLLECTED		3,000.00	**OUT** · money	USDT	client	principal	
CL.ALI.EUR.DUE	2,646.00		**IN** · money	EUR	client	principal	client net is unchanged: the sender's price is write-once
CO.ALI.EUR.POOL	12.00		**IN** · money	EUR	company	earnings	margin actually left at Ali
CO.NONE.EUR.VAR_RATE_HONOUR	15.00		**IN** · cost	EUR	company	variance	the reused rate vs the day's rate: NPL's cost, chosen
CL.PROJECT.EUR.PAYABLE		2,646.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		27.00	**OUT** · earned	EUR	company	earnings	margin as priced

Pattern L — Loss events: booked to NPL first, recovered share by share

When money is lost after it was verified (a frozen account, cash gone missing), NPL takes the whole loss on itself at once and makes the client whole. Then, as each party agrees to bear a share, that share is recovered.

T39 · Deal 7 collected: 10,000 USDT verified on a single-use local account at Ali (shown as T1)

source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		10,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

A 10,000 USDT deal is paid into a one-time local account at Ali and verified. Normal collection lines.

T40 · Loss event: the account is frozen before conversion and the 10,000.00 USDT are lost; NPL books the whole loss and makes the client whole

source: LossEvent

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.USDT.EXP_LOSS	10,000.00		**IN** · cost	USDT	company	loss	the whole loss is NPL's first (LOSS_EVENT state approved → posted)
CO.ALI.USDT.POOL		6,000.00	**OUT** · money	USDT	company	loss	NPL's money at Ali funds the make-good as far as it goes (LOSS_EVENT.pool_funded)
CO.OWN.USDT.WALLET		4,000.00	**OUT** · money	USDT	company	loss	the rest is transferred from NPL's own wallet to Ali (LOSS_EVENT.wallet_funded, funding_reference)
CL.ALI.USDT.COLLECTED		10,000.00	**OUT** · money	USDT	client	principal	the client money that was lost
CL.ALI.USDT.COLLECTED	10,000.00		**IN** · money	USDT	client	loss	replaced by NPL's money: the deal proceeds, the collection part stays verified

WHAT IS HAPPENING

Then the account is frozen and the money is lost. NPL's rule: book the whole loss to NPL first and make the client whole immediately. So 10,000.00 goes IN to NPL's loss jar; NPL funds the replacement 6,000.00 from its pool at Ali and 4,000.00 from its own wallet (physically transferred to Ali); the lost client money leaves Ali's jar and the replacement enters it. The client's promise never changes and the deal carries on. A partial loss works the same way for the lost part only: if 4,000 of the 10,000 were lost, every line reads 4,000 and the other 6,000 convert normally.

THE ARITHMETIC

- lost 10,000.00 → NPL's loss jar; replaced from NPL's pool at Ali 6,000.00 + NPL's wallet 4,000.00

T41a · Management approves the split (approve_loss_split): partner 4,000.00, sender 2,000.00, client 1,000.00, NPL 3,000.00 USDT — Ali's share agreed (LOSS_RECOVERY party partner, state agreed) source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.USDT.LOSS_RECEIVABLE	4,000.00		**IN** · money	USDT	company	loss	Ali's agreed share, now owed to NPL
CO.NONE.USDT.EXP_LOSS		4,000.00	**OUT** · cost	USDT	company	loss	NPL's expense falls by Ali's share

WHAT IS HAPPENING

Management then decides who bears what: Ali 4,000.00, the sender 2,000.00, the client 1,000.00, NPL 3,000.00. Each share is its own recovery record and its own posting, all three made at the moment of approval. First Ali's: its share becomes a receivable (a promise Ali owes NPL) and comes off the loss jar.

THE ARITHMETIC

- split: Ali 4,000.00 + sender 2,000.00 + client 1,000.00 + NPL 3,000.00 = 10,000.00

T41b · The sender's share agreed: 2,000.00 USDT (LOSS_RECOVERY party sender, state agreed) source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.SENDER_A.USDT.LOSS_RECEIVABLE	2,000.00		**IN** · money	USDT	company	loss	the sender's agreed share, owed to NPL
CO.NONE.USDT.EXP_LOSS		2,000.00	**OUT** · cost	USDT	company	loss	NPL's expense falls by the sender's share

WHAT IS HAPPENING

The sender's share, the same way: a receivable from the sender, off the loss jar.

T41c · The client's share: 1,000.00 USDT borne by the client, received at once by partial reversal of the make-good (LOSS_RECOVERY party client, state received) source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.USDT.POOL	1,000.00		**IN** · money	USDT	company	loss	NPL takes back the part of its make-good the client bears
CO.NONE.USDT.EXP_LOSS		1,000.00	**OUT** · cost	USDT	company	loss	NPL's expense falls to its own share, 3,000.00
CL.PROJECT.USDT.PAYABLE	1,000.00		**IN** · promise	USDT	client	loss	the client's claim falls by the share it bears
CL.ALI.USDT.COLLECTED		1,000.00	**OUT** · money	USDT	client	loss	and the restored balance with it

WHAT IS HAPPENING

The client's share is handled by undoing that part of the make-good: the client's promise and its jar at Ali both drop by 1,000.00, and NPL takes that much back into its pool. Nothing is owed by anyone, so this recovery counts as received at once. The loss jar ends at NPL's own share, 3,000.00.

THE ARITHMETIC

- loss jar: 10,000.00 − 7,000.00 recovered = 3,000.00 left, NPL's own share

T42 · Ali settles its share: 4,000.00 USDT credited to NPL's pool at Ali source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.USDT.POOL	4,000.00		**IN** · money	USDT	company	loss	received: the LOSS_RECOVERY row of T41a moves to state received
CO.ALI.USDT.LOSS_RECEIVABLE		4,000.00	**OUT** · money	USDT	company	loss	

WHAT IS HAPPENING

Ali pays its share by crediting NPL's pool at Ali. Receivable cleared.

T43 · The sender settles its share: 2,000.00 USDT paid to NPL's own wallet source: LossRecovery

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.OWN.USDT.WALLET	2,000.00		**IN** · money	USDT	company	loss	a payment to NPL, never a netting against a sender credit (client money); the LOSS_RECOVERY row of T41b moves to state received
CO.SENDER_A.USDT.LOSS_RECEIVABLE		2,000.00	**OUT** · money	USDT	company	loss	

WHAT IS HAPPENING

The sender pays its share into NPL's wallet. Receivable cleared. (A sender's credit from an overpayment is never used for this: that is client money.)

Pattern M — The reseller project (NPL-GR): two invoices, one remittance deal

NPL-GR buys goods from a vendor and resells them. Its profit is the difference between the vendor's invoice and its own resale invoice, both in euros, fixed up front. The customer pays through NPL's remittance service, which charges its own 2 % and absorbs all the currency ups and downs so the reseller profit is never touched.

T44 · NPL-GR receives the vendor's invoice: 9,000.00 EUR at NPL's preferential price (INVOICE kind vendor)

source: Invoice

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.EUR.RESELL_COST	9,000.00		**IN** · cost	EUR	company	resale	NPL-GR's cost, in the invoice currency, when received
CO.VENDOR_V.EUR.VENDOR_PAYABLE		9,000.00	**OUT** · promise	EUR	company	resale	owed to the vendor; INVOICE_BALANCE shows the open part

WHAT IS HAPPENING

NPL-GR, NPL's game-reseller unit, receives a vendor's invoice for 9,000.00 EUR. These two lines are NPL-GR's own books, kept in euros: cost IN (what the goods cost) and a promise OUT to the vendor (what is owed). Nothing has moved yet.

THE ARITHMETIC

- vendor invoice 9,000.00 EUR → cost and promise to the vendor

T45 · NPL-GR issues the resale invoice to its customer: 9,450.00 EUR at retail price (INVOICE kind resale); game reseller revenue 450.00 EUR is fixed here

source: Invoice

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.CUSTOMER_C.EUR.RESALE_RECEIVABLE	9,450.00		**IN** · money	EUR	company	resale	owed by the customer
CO.NONE.EUR.RESELL_REVENUE		9,450.00	**OUT** · earned	EUR	company	resale	NPL-GR's revenue, in the invoice currency, when issued

WHAT IS HAPPENING

NPL-GR sells the goods on at 9,450.00 EUR and issues its own invoice. Again in its own books: a promise IN from the customer (a receivable) and revenue earned. The difference, 450.00, is the reseller profit, and it is now fixed. Nothing that follows may change it.

THE ARITHMETIC

- resale invoice 9,450.00 EUR → revenue and the customer's promise; reseller profit 9,450.00 − 9,000.00 = 450.00, fixed now

T46 · The customer pays the resale invoice in THB cash to Ali: 359,746.78 THB = 9,450.00 EUR at the sender rate with the 2 % remittance markup, rounded up to 360,000.00 THB in cash (cash_rounding, unit 1,000) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.THB. COLLECTED	360,000.00		**IN** · money	THB	client	principal	client money at Ali — the client of this remittance project is NPL-GR
CL.PROJECT.THB. PAYABLE		360,000.00	**OUT** · promise	THB	client	principal	the remittance side owes it onward on NPL-GR's behalf
CO.PROJECT.EUR. REMIT_CLAIM	9,450.00		**IN** · money	EUR	company	resale	reseller books: the customer has paid, the remittance side now holds the value for NPL-GR
CO.CUSTOMER_C.EUR. RESALE_RECEIVABLE		9,450.00	**OUT** · money	EUR	company	resale	the customer's debt is settled

WHAT IS HAPPENING

The customer pays the 9,450.00 EUR invoice in Thai baht cash through NPL's remittance service, with the usual 2 % remittance fee on top. The exact figure would be 359,746.78 THB (the invoice's USDT equivalent, plus 2 %, at the market rate); cash is rounded up to the thousand, so 360,000.00 THB are handed over and counted at Ali. The first two lines are an ordinary collection (NPL-GR is simply the client here). The last two are NPL-GR's own books noting that the customer has paid and the remittance side now holds the value. The few baht of rounding surplus are NPL-GR's and come out at the end (T51).

THE ARITHMETIC

- USDT needed for 9,450.00 EUR at 0.8708 = 10,852.09
- THB to collect = 10,852.09 × (1 + 2 %) × 32.50 THB per USDT (market) = 359,746.78 THB — in target mode the fee is a markup on the target
- cash rounded up to the 1,000: 360,000.00 THB handed over; the 253.22 THB surplus is NPL-GR's

WHAT IS HAPPENING

Ali converts baht to USDT. Just like T2: the client's share (10,859.73 USDT, worth exactly 9,450.00 EUR at the day's rate) goes into the client's jar, and NPL's remittance fee less Ali's cost (106.42) into NPL's pool. The reseller profit is not touched; it lives in euros in T45.

THE ARITHMETIC

- 360,000.00 THB at market = 11,076.92 USDT; NPL's 2 % markup = 217.19; client net 10,859.73 (= 9,450.00 EUR at 0.8708 plus the 7.63 USDT cash surplus)
- Ali converts at 32.83: 10,966.15 delivered; NPL's margin 106.42 (2 % less Ali's 1 %) → pool

T47 · Conversion at Ali (fee_in_rate, markup on market): 360,000.00 THB → 10,966.15 USDT at 32.83 (market 32.50); client net 10,859.73 USDT (= 10,852.09 for the 9,450.00 EUR at the group's invoice_rate 1.148369 USDT per EUR, plus the 7.63 USDT cash-rounding surplus), NPL's earnings 106.42 USDT source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.THB.PAYABLE	360,000.00		**IN** · promise	THB	client	principal	
CL.ALI.THB.COLLECTED		360,000.00	**OUT** · money	THB	client	principal	
CL.ALI.USDT.DUE	10,859.73		**IN** · money	USDT	client	principal	NPL-GR's entitlement in USDT, reserved for the vendor and for NPL-GR's margin
CO.ALI.USDT.POOL	106.42		**IN** · money	USDT	company	earnings	remittance earnings resting at Ali
CL.PROJECT.USDT.PAYABLE		10,859.73	**OUT** · promise	USDT	client	principal	
CO.NONE.USDT.EARN_GROSS		106.42	**OUT** · earned	USDT	company	earnings	0.96 % of the market value of the THB: the 2 % markup on the target (217.19) less Ali's 1 % inside its rate (110.77)

T48 · Instalment 1 paid by Ali: 3,445.11 USDT to the vendor, discharging 3,000.00 EUR of the vendor invoice at discharge_rate 1.148369 USDT per EUR (release; the vendor's confirmation posts nothing more) source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	3,445.11		**IN** · promise	USDT	client	principal	released by Ali on NPL's instruction: the promise is discharged
CL.ALI.USDT.DUE		3,445.11	**OUT** · money	USDT	client	principal	
CO.VENDOR_V.EUR.VENDOR_PAYABLE	3,000.00		**IN** · promise	EUR	company	resale	reseller books: the vendor is owed 3,000 EUR less (vendor INVOICE_BALANCE 6,000, counted over released lines)
CO.PROJECT.EUR.REMIT_CLAIM		3,000.00	**OUT** · money	EUR	company	resale	the remittance side holds 3,000 EUR less for NPL-GR

WHAT IS HAPPENING

First instalment to the vendor: 3,445.11 USDT pays off 3,000.00 EUR of the vendor invoice at today's rate. Remittance side: the money leaves the client's jar at Ali and the promise is discharged on release, as in T7; the vendor's confirmation posts nothing more. NPL-GR's books: the vendor is owed 3,000.00 less and the remittance side holds 3,000.00 less for NPL-GR.

THE ARITHMETIC

- 3,000.00 EUR ÷ 0.8708 = 3,445.11 USDT paid, same rate as the quote day, so no difference

T49 · Variant route: the remaining 7,414.62 USDT of NPL-GR's balance hop from Ali to NPL's own desk (LT Sub at Aquanow; a SETTLEMENT_LINE to the partner_transit receiver, posted when the own desk credits) source: Confirmation

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWNDESK.USDT.DUE	7,414.62		**IN** · money	USDT	client	principal	credited at the own desk: an ordinary partner-coded account with the own-desk configuration as holder, reported by the custody view
CL.ALI.USDT.DUE		7,414.62	**OUT** · money	USDT	client	principal	left Ali; a hop is not a delivery, so the promise is untouched

WHAT IS HAPPENING

A variant of where the money sits: instead of Ali paying the vendor, the remaining 7,414.62 USDT are moved to NPL's own desk (its wallet at Aquanow), which will pay. It posts once, when the own desk credits the money: OUT of the Ali jar, IN to the own-desk jar, which is just another "at a partner" jar with NPL as the partner. A hop is not a delivery, so the promise is untouched.

THE ARITHMETIC

- client USDT left at Ali: 10,859.73 − 3,445.11 = 7,414.62 → moved to the own desk

WHAT IS HAPPENING

Four days later the second instalment: the rate has moved, so the remaining 6,000.00 EUR costs 6,855.03 USDT instead of the 6,890.22 set aside. The 35.19 difference is a currency gain, and it belongs to the remittance side, not to NPL-GR: so it moves from the client's jar to NPL's pool and is recorded in a "fx timing" jar. Had the rate gone the other way, NPL's pool would have topped the client's jar up. The promise is discharged on release, as always. NPL-GR's books: vendor fully paid.

THE ARITHMETIC

- 6,000.00 EUR at the quote-day rate 0.8708 would be 6,890.22 USDT; at today's 0.87527 it is 6,855.03; gain 35.19 → from the client's jar to NPL's pool, recorded as fx timing

T50 · Instalment 2 paid by the own desk four days later: 6,855.03 USDT discharge the remaining 6,000.00 EUR at discharge_rate 1.142505 USDT per EUR; 6,890.22 USDT had been reserved at the group's invoice_rate 1.148369, so 35.19 USDT are a forex gain for the remittance side source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	6,855.03		**IN** · promise	USDT	client	principal	released: the promise is discharged
CL.OWNDESK.USDT.DUE		6,855.03	**OUT** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE	35.19		**IN** · promise	USDT	client	fx_timing	forex true-up: the 6,000 EUR cost fewer USDT than reserved, the surplus is not NPL-GR's
CL.OWNDESK.USDT.DUE		35.19	**OUT** · money	USDT	client	fx_timing	it leaves NPL-GR's balance
CO.OWNDESK.USDT.POOL	35.19		**IN** · money	USDT	company	fx_timing	and becomes NPL's, resting at the own desk
CO.NONE.USDT.VAR_FX_TIMING		35.19	**OUT** · cost	USDT	company	fx_timing	a gain; with the rate the other way the line is a debit and the pool tops NPL-GR's balance up
CO.VENDOR_V.EUR.VENDOR_PAYABLE	6,000.00		**IN** · promise	EUR	company	resale	reseller books: the vendor invoice is settled (INVOICE_BALANCE 0)
CO.PROJECT.EUR.REMIT_CLAIM		6,000.00	**OUT** · money	EUR	company	resale	

WHAT IS HAPPENING

What is left in the client's jar is NPL-GR's reseller profit, 450.00 EUR worth, plus the 7.63 USDT of cash-rounding surplus from T46; all of it is paid to NPL-GR's wallet, 521.76 USDT (with a small fx gain again on the profit part). Because NPL-GR's books count in euros but the money arrived in USDT, a "currency result" pair records the exchange: euros sold out of the claim, USDT bought into the wallet. The small cash-rounding surplus is kept apart as extra reseller revenue in USDT, since it has nothing to do with exchange rates. The two invoices are now fully settled and the 450.00 profit is intact.

THE ARITHMETIC

- profit 450.00 EUR at 0.87527 = 514.13 USDT; set aside at 0.8708 was 516.77; gain 2.64
- plus the cash-rounding surplus 7.63 USDT: paid out 521.76 in all
- FX_RESULT pair: 450.00 EUR sold out of NPL-GR's claim, 514.13 USDT bought into NPL's wallet; the 7.63 USDT surplus goes straight to reseller revenue

T51 · The margin line: 450.00 EUR of the resale invoice (514.13 USDT at 1.142505 USDT per EUR) plus the 7.63 USDT cash-rounding surplus are paid to NPL-GR's own wallet, 521.76 USDT in all; 516.77 USDT had been reserved for the margin, 2.64 USDT forex gain source: Settlement

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	521.76		**IN** · promise	USDT	client	principal	released to NPL-GR's wallet (a receiver of the project); the balance is empty. The resale invoice's own balance went to zero at T46, when the group that collects its whole amount was paid
CL.OWNDESK.USDT.DUE		521.76	**OUT** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE	2.64		**IN** · promise	USDT	client	fx_timing	forex true-up on the margin line
CL.OWNDESK.USDT.DUE		2.64	**OUT** · money	USDT	client	fx_timing	
CO.OWNDESK.USDT.POOL	2.64		**IN** · money	USDT	company	fx_timing	
CO.NONE.USDT.VAR_FX_TIMING		2.64	**OUT** · cost	USDT	company	fx_timing	
CO.OWN.USDT.WALLET	521.76		**IN** · money	USDT	company	resale	reseller books: NPL-GR's margin arrives, in USDT
CO.NONE.USDT.FX_RESULT		514.13	**OUT** · cost	USDT	company	resale	currency bought: the USDT received for the 450 EUR claim at the day's rate
CO.NONE.USDT.RESELL_REVENUE		7.63	**OUT** · earned	USDT	company	rounding	the cash-rounding surplus is NPL-GR's revenue in the currency received, never a forex result
CO.NONE.EUR.FX_RESULT	450.00		**IN** · cost	EUR	company	resale	currency sold: the EUR claim given up
CO.PROJECT.EUR.REMIT_CLAIM		450.00	**OUT** · money	EUR	company	resale	nothing is held for NPL-GR any more

Pattern N — Earnings kept in the unconverted balance and offset between partners

Partners convert everything they receive, so NPL's fee normally ends up converted along with the client's money. NPL prefers to keep its fees in USDT, taken out of money that has not been converted yet, even when the fee was earned at a different partner. The ledger records that as a swap between NPL and the client at the day's rate.

T52a · Collection at Jeton: 300,000 USDT (as T1) source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.JETON.USDT.COLLECTED	300,000.00		**IN** · money	USDT	client	principal	
CL.PROJECT.USDT.PAYABLE		300,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

A large collection: 300,000 USDT arrive at Jeton. Money IN, promise OUT.

WHAT IS HAPPENING

Jeton converts all of it. The client gets 264,600.00 EUR (3,780.00 sender fee and 1,620.00 receiver fee came off the 270,000.00 at market). Jeton keeps 3,375.00 as its fee (NPL's cost), and NPL's margin of 2,025.00 EUR sits in its pool at Jeton, in euros, mixed into Jeton's balance where Jeton cannot see it.

THE ARITHMETIC

- 300,000.00 USDT × market 0.9000 = 270,000.00 EUR at market
- fee 2.00 % of that = 5,400.00 (sender pays 1.40 % = 3,780.00, receiver pays 0.60 % = 1,620.00)
- client net = 270,000.00 − 5,400.00 = 264,600.00 EUR → the client's jar and promise
- Jeton converts at market: 270,000.00, then charges 1.25 % = 3,375.00 → NPL's cost jar; delivered 266,625.00
- NPL's margin = delivered 266,625.00 − client net 264,600.00 = 2,025.00 → NPL's pool at Jeton; fee charged 5,400.00 − partner cost 3,375.00 = 2,025.00

T52 · Jeton converts the 300,000 USDT in full → 266,625.00 EUR; NPL's margin 2,025.00 EUR now sits at Jeton in EUR (the collection posted as in T1 and is not repeated) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	300,000.00		**IN** · promise	USDT	client	principal	the USDT promise is discharged by the conversion
CL.JETON.USDT.COLLECTED		300,000.00	**OUT** · money	USDT	client	principal	
CL.JETON.EUR.DUE	264,600.00		**IN** · money	EUR	client	principal	client net (dust ignored here)
CO.JETON.EUR.POOL	2,025.00		**IN** · money	EUR	company	earnings	NPL's margin, in EUR, inside Jeton's balance; Jeton does not know it is there
CO.NONE.EUR.EXP_PARTNER	3,375.00		**IN** · cost	EUR	company	partner_cost	Jeton's stated 1.25 %
CL.PROJECT.EUR.PAYABLE		264,600.00	**OUT** · promise	EUR	client	principal	
CO.NONE.EUR.EARN_GROSS		5,400.00	**OUT** · earned	EUR	company	earnings	

T53 · Meanwhile 100,000 USDT arrive at Ali for another deal, scheduled to convert the next business day source: Collection

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.USDT.COLLECTED	100,000.00		**IN** · money	USDT	client	principal	unconverted at Ali
CL.PROJECT.USDT.PAYABLE		100,000.00	**OUT** · promise	USDT	client	principal	

WHAT IS HAPPENING

Meanwhile 100,000 USDT arrive at Ali for another deal that will convert tomorrow. Money IN, promise OUT, nothing converted yet.

WHAT IS HAPPENING

NPL would rather hold its margin in USDT at Ali than in euros at Jeton. So it swaps: its 2,025.00 EUR at Jeton becomes the client's (client jar at Jeton up, NPL pool at Jeton down), and in exchange 2,272.73 USDT of the client's unconverted money at Ali becomes NPL's (client jar at Ali down, NPL pool at Ali up). The client's promises are adjusted the same way: fewer USDT owed, more euros owed. For the client it is as if 2,272.73 USDT had been converted at 0.891; its total value is unchanged. For NPL it is an exchange of euros for USDT, and the "currency result" pair records euros sold and USDT bought. Nothing moves physically; both partners see nothing.

THE ARITHMETIC

- NPL's margin at Jeton 2,025.00 EUR ÷ Ali's rate 0.891 = 2,272.73 USDT: that much of the client's USDT at Ali becomes NPL's, and 2,025.00 EUR at Jeton becomes the client's

T54 · Earnings offset between partners (EARNINGS_OFFSET): NPL's 2,025.00 EUR at Jeton become client money there; 2,272.73 USDT of the client's unconverted balance at Ali (= 2,025.00 EUR at Ali's rate 0.891) become NPL's source: EarningsOffset

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.JETON.EUR.DUE	2,025.00		**IN** · money	EUR	client	earnings	Jeton's EUR balance is now entirely client money
CL.PROJECT.EUR.PAYABLE		2,025.00	**OUT** · promise	EUR	client	earnings	the client's EUR claim grows by the same amount
CL.PROJECT.USDT.PAYABLE	2,272.73		**IN** · promise	USDT	client	earnings	and its USDT claim falls: for the client this is a conversion of USDT at Ali into EUR at Jeton, at the day's rate
CL.ALI.USDT.COLLECTED		2,272.73	**OUT** · money	USDT	client	earnings	that much USDT at Ali is no longer the client's
CO.ALI.USDT.POOL	2,272.73		**IN** · money	USDT	company	earnings	NPL's margin now rests at Ali, in USDT, inside the unconverted balance
CO.NONE.USDT.FX_RESULT		2,272.73	**OUT** · cost	USDT	company	earnings	currency bought: the USDT NPL took at Ali
CO.NONE.EUR.FX_RESULT	2,025.00		**IN** · cost	EUR	company	earnings	currency sold: the EUR NPL gave up at Jeton
CO.JETON.EUR.POOL		2,025.00	**OUT** · money	EUR	company	earnings	nothing of NPL's rests at Jeton any more; the EARNINGS_RECEIVABLE rows at Jeton are recovered by this offset

WHAT IS HAPPENING

The next day Ali converts, but only 96,717.17 USDT: NPL also leaves its own margin on this deal, 1,010.10 USDT, unconverted and takes it into its USDT pool (earned in USDT this time). The client receives 86,175.00 EUR at Ali, which together with the 2,025.00 it already holds at Jeton is exactly the 88,200.00 it was quoted. End state: Jeton's euros are all the client's; Ali's USDT are all NPL's; Ali's euros are all the client's.

THE ARITHMETIC

- Ali deal: 100,000 × 0.9000 = 90,000.00; fee 2 % = 1,800.00; client net 88,200.00; Ali at 0.891 delivers 89,100.00; margin 900.00 EUR
- margin kept in USDT: 900.00 ÷ 0.891 = 1,010.10
- USDT converted = 100,000 − 2,272.73 (offset) − 1,010.10 (kept) = 96,717.17; × 0.891 = 86,175.00 EUR to the client at Ali
- client total: 86,175.00 at Ali + 2,025.00 at Jeton = 88,200.00, as quoted; NPL's USDT at Ali: 2,272.73 + 1,010.10 = 3,282.83

T55 · Ali converts the next day, less NPL's margins: 96,717.17 USDT → 86,175.00 EUR; the Ali deal's own margin 900.00 EUR is kept as 1,010.10 USDT instead (CONVERSION.residual_in) source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	97,727.27		**IN** · promise	USDT	client	principal	the client's remaining USDT claim at Ali
CL.ALI.USDT.COLLECTED		96,717.17	**OUT** · money	USDT	client	principal	what Ali actually converts
CL.ALI.USDT.COLLECTED		1,010.10	**OUT** · money	USDT	client	earnings	NPL's margin on this deal, left unconverted
CO.ALI.USDT.POOL	1,010.10		**IN** · money	USDT	company	earnings	and recognised in USDT
CO.NONE.USDT.EARN_GROSS		1,010.10	**OUT** · earned	USDT	company	earnings	= 900.00 EUR at 0.891: the margin is earned in the collection currency
CL.ALI.EUR.DUE	86,175.00		**IN** · money	EUR	client	principal	client net less the part already delivered at Jeton
CL.PROJECT.EUR.PAYABLE		86,175.00	**OUT** · promise	EUR	client	principal	the client's EUR entitlement on this deal (write-once) less what it already holds at Jeton

Pattern O — reroute after conversion

Sometimes a partner has already turned the client's money into euros and then cannot pay them out: it is suspended, its bank is down, a limit is hit, or it is a holiday. NPL decided on 20 October that in that case the money does not simply wait. It is moved to another partner: the first partner turns the euros back into USDT at its rate of the day, sends the USDT to NPL's own wallet, NPL forwards it to the second partner, and the second partner turns it into euros again at its own rate. The client is promised exactly what it was promised before; whatever the round trip costs or gains is NPL's. Management must approve, because for a while NPL's own wallet is holding the client's money, which is the fourth of the cases in which that is allowed.

WHAT IS HAPPENING (T56). Ali has 8,820.00 EUR of the client's money from deal 1 and cannot pay it out. Ali converts it back: the euro promise is torn up (IN · promise) and the euros leave Ali's jar; 9,834.30 USDT arrive in the client's USDT jar at Ali with a USDT promise of the same amount. No fee: this is not a new deal, only a move. THE ARITHMETIC: 8,820.00 EUR × 1.1150 USDT per EUR = 9,834.30 USDT (the 1.1150 is illustrative).

T56 · Ali converts the client's 8,820.00 EUR balance back to 9,834.30 USDT at 1.1150 (BALANCE_CONVERSION owned by the REROUTE, phase post_conversion; no markup) — source: BalanceConversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.EUR.PAYABLE	8,820.00		IN · promise	EUR	client	principal	the EUR entitlement is re-stated in USDT for the move; the client net as priced is remembered on the reroute
CL.ALI.EUR.DUE		8,820.00	OUT · money	EUR	client	principal	the converted balance leaves the client's EUR jar at Ali
CL.ALI.USDT.DUE	9,834.30		IN · money	USDT	client	principal	8,820.00 × 1.1150 = 9,834.30 USDT at Ali's EUR → USDT rate of the day, no markup
CL.PROJECT.USDT.PAYABLE		9,834.30	OUT · promise	USDT	client	principal	the same obligation, back in the collection currency

WHAT IS HAPPENING (T57). Ali sends the USDT to NPL's own wallet: the money goes IN to the “held in NPL's wallet” jar and OUT of the client's jar at Ali. Then NPL forwards it to Jeton: OUT of the held jar, IN to the client's USDT jar at Jeton. The promise does not change at either step, because nothing has been delivered. If the network charges a fee for the forward, NPL pays it as in T17.

T57 · Ali pays 9,834.30 USDT to NPL's contingency wallet and NPL forwards it to Jeton (two outbound transfers under Section 5; forward_full) — source: Reroute

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.OWN.USDT.HELD	9,834.30		IN · money	USDT	client	principal	Ali pays the USDT to NPL's contingency wallet; client money in custody with deal, reason and age (custody case: reroute after conversion)
CL.ALI.USDT.DUE		9,834.30	OUT · money	USDT	client	principal	nothing of the client's is left at Ali for this deal
CL.JETON.USDT.COLLECTED	9,834.30		IN · money	USDT	client	principal	NPL forwards the full amount to Jeton (forward_full; a network fee, when one applies, posts as in T17)
CL.OWN.USDT.HELD		9,834.30	OUT · money	USDT	client	principal	custody ends

WHAT IS HAPPENING (T58). Jeton converts at the market rate and charges its own fee separately. 9,834.30 USDT × 0.9000 = 8,850.87 EUR; Jeton keeps 1.25 % = 110.64 and delivers 8,740.23. But the client was promised 8,820.00, and that promise never changes, so 8,820.00 goes IN to the client's euro jar at Jeton and the promise is re-stated in euros. The 79.77 Jeton delivered short of that is NPL's: Jeton's fee (110.64, into NPL's partner-cost jar) less a small gain because 8,850.87 at market was more than the 8,820.00 the client is owed (30.87, into the variance jar as a gain). NPL's pool at Jeton goes 79.77 below zero, exactly as after the bounce fee in T24, and is squared at the next conversion there (T24a). NPL's 90.00 margin from the first conversion is still sitting in its pool at Ali; a reroute never claws back what was earned.

T58 · Jeton converts 9,834.30 USDT at market 0.9000 = 8,850.87 EUR, keeps 1.25 % = 110.64, delivers 8,740.23; the client net 8,820.00 is unchanged — source: Conversion

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.PROJECT.USDT.PAYABLE	9,834.30		IN · promise	USDT	client	principal	the USDT obligation is discharged by the second conversion
CL.JETON.USDT.COLLECTED		9,834.30	OUT · money	USDT	client	principal	
CL.JETON.EUR.DUE	8,820.00		IN · money	EUR	client	principal	the client net as priced on the first conversion: write-once, so the client is exactly where it was
CL.PROJECT.EUR.PAYABLE		8,820.00	OUT · promise	EUR	client	principal	owed to the receiver group, in EUR again
CO.NONE.EUR.EXP_PARTNER	110.64		IN · cost	EUR	company	partner_cost	Jeton's stated 1.25 % of the 8,850.87 it converted at market; NPL's cost, since no fee is earned on a reroute
CO.NONE.EUR.VAR_CONVERSION		30.87	OUT · cost	EUR	company	variance	basis variance on the reroute: 8,850.87 converted at market less the 8,820.00 client net, a gain this time (a loss would be a debit)
CO.JETON.EUR.POOL		79.77	OUT · money	EUR	company	earnings	Jeton delivered 8,740.23 against a client jar of 8,820.00: NPL's pool at Jeton goes to −79.77 until squared as in T24a

Balances after Pattern O

Account	Ccy	Balance	Meaning
CL.JETON.EUR.DUE	EUR	8,820.00	client balance, now at Jeton, as priced
CL.PROJECT.EUR.PAYABLE	EUR	8,820.00	owed to receivers: unchanged by the move
CL.ALI.EUR.DUE / CL.ALI.USDT.DUE	EUR / USDT	0.00	nothing of the client's left at Ali for this deal
CO.ALI.EUR.POOL	EUR	90.00	NPL's margin from the first conversion, untouched
CO.JETON.EUR.POOL	EUR	−79.77	the cost of the move, squared at the next conversion at Jeton (T24a)
CO.NONE.EUR.EXP_PARTNER	EUR	110.64	Jeton's stated fee on the second conversion
CO.NONE.EUR.VAR_CONVERSION	EUR	−30.87	basis variance on the reroute (negative = gain)

Pattern P — Interruptions between instruction and recording

What happens when something breaks in the middle: the partner has done its part but NPLify did not record it, a receiver never says the money arrived, or the blockchain explorer is down. The rule that keeps the books safe is simple: no automatic process ever moves value between jars. A person records what happened, once, with the evidence, and the system refuses to record the same event twice. So a breakdown can make a posting late; it can never make it double. The three cases and how each is resolved are set out in Pattern P of the Ledger Posting Design; the worked postings are added once the real behaviour has been exercised on staging.

Pattern Q — Month-end reconciliation with a partner statement

Once a month, and every evening during the parallel run, Finance puts the partner's statement next to the ledger. For each partner and currency the partner holds one balance; NPL's books say the same money is partly the client's jar and partly NPL's pool jar, and the two jars must add up to the partner's figure. When they do not, Finance finds out why before anything is posted: a payout the partner has already sent but the receiver has not yet confirmed (a timing difference, nothing to post), a charge the partner took and did not advise (a true-up, below), or a mistake on the partner's side (a query to the partner, nothing to post). Each reconciliation is kept as a record with the figures, the cause and who resolved it.

WHAT IS HAPPENING (T59). The ledger says Ali holds 8,910.00 EUR for NPL: 8,820.00 of the client's (deal 1) and 90.00 of NPL's fee. Ali's statement says 8,905.00. The detail shows a 5.00 transfer charge Ali took on the day. NPL bears it: 5.00 goes IN to NPL's bank-fee cost jar and OUT of NPL's pool at Ali. The client's jar is untouched, because the charge was Ali's, not the receiver's. Now NPL's books say 8,820.00 + 85.00 = 8,905.00, the same as Ali. THE ARITHMETIC: 8,820.00 + 90.00 = 8,910.00; statement 8,905.00; difference 5.00 = the unadvised charge; pool 90.00 − 5.00 = 85.00.

T59 · Month-end true-up at Ali: the statement shows 8,905.00 EUR, the ledger 8,910.00 (client 8,820.00 + pool 90.00); the 5.00 difference is a charge not yet booked — source: Adjustment

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.NONE.EUR.EXP_BANKFEE	5.00		IN · cost	EUR	company	bank_fee	Ali's statement shows 5.00 EUR less than the ledger: a transfer charge Ali deducted on the day and had not advised; NPL bears it (treatment absorbed)
CO.ALI.EUR.POOL		5.00	OUT · money	EUR	company	bank_fee	funded from NPL's pool at Ali; the client's balance is untouched because the charge was Ali's, not the receiver's

Pattern R — Opening entries on the opening date

On the day NPLify takes over, the money is already where it is: the client's euros are waiting at Ali, NPL's fees are resting there too, a rebate is owed by Jeton, a sender has a credit. Rather than re-typing months of old deals to arrive at those figures, Finance lists every such balance from the sheets, checks each one against the partner's statement or the wallet, Management approves the list, and each line is posted once as an opening entry with the statement attached. The jars start full, and everything that happens afterwards is an ordinary posting on top.

WHAT IS HAPPENING (T60). The client's 8,820.00 EUR waiting at Ali goes IN to the client's euro jar at Ali, and the promise to deliver it goes OUT, exactly the shape the conversion T2 would have left; the receiver group is on the promise line, so the entitlement view is right from the first day. No NPL account is involved: it is the client's money and the client's promise.

T60 · Opening entry, client balance: 8,820.00 EUR of the client's money waiting at Ali on the opening date — source: Opening

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CL.ALI.EUR.DUE	8,820.00		IN · money	EUR	client	principal	client money converted and waiting at Ali on the opening date, per the sheet and Ali's statement
CL.PROJECT.EUR.PAYABLE		8,820.00	OUT · promise	EUR	client	principal	owed to the receiver group (receiver_group_id set); split by sender on the OPENING_ENTRY rows

WHAT IS HAPPENING (T61). NPL's own 90.00 EUR at Ali goes IN to NPL's pool jar at Ali, and OUT of a jar used only on this day: “what NPL brought in”. That jar is never touched again; its total is simply the starting point of NPL's own money. Now Ali's one balance of 8,910.00 is the client's 8,820.00 plus NPL's 90.00, as the identity requires. A balance that NPL is owed (a rebate not yet paid) or owes (a markup share) opens the same way against that one jar.

T61 · Opening entry, NPL's pool: 90.00 EUR of NPL's own money resting at Ali on the opening date — source: Opening

Account	Debit	Credit	In / Out	Ccy	Owner	Component	Why
CO.ALI.EUR.POOL	90.00		IN · money	EUR	company	earnings	NPL's margin resting at Ali on the opening date, verified against Ali's statement (one balance of 8,910.00 = 8,820.00 client + 90.00 NPL)
CO.NONE.EUR.OPENING_BALANCE		90.00	OUT · equity	EUR	company	earnings	what NPL brought in; posted once

Pattern G — inter-group offsets, and what is not here

One pattern of the main document has no table because it has no special lines: when a receiver of one group is paid out of money collected for another group, the payout posts exactly like Pattern C, and a report (not an account) shows one group owed and the other holding its flow. The main document also carries the full chart of accounts, the identities that must hold after every posting, the reports built from the ledger, and who may post what. This companion is for reading the examples; that one is the reference.

Figures are worked examples, not NPL data, except where the main document says otherwise.

Appendix · Identifier register

Every worked transaction, acceptance vector, walkthrough, flow and common procedure in the pack has one identifier, listed here so that a requirement, a fixture in the test suite and a line in a review can name the same thing. The identifiers are frozen from this edition: none is renumbered, none is reused, a new item takes the next free number, and a withdrawn item keeps its number with the note “withdrawn”. The 61 transactions of this companion (T1 to T55 with their a, b and c variants) are the fixtures of the automated ledger tests from Phase 1 (M2); T56 to T59 join them as they are written.

Identifiers	What they name	Where
T1, T2, T3a, T3	Pattern A — direct collection, conversion and accumulation	Ledger Posting Design §6; this companion
T4 to T6	Pattern B — part collection	§7
T7 to T11	Pattern C — settlement, confirmation, bank fees, shortfalls, rounding	§8
T12 to T17	Pattern D — contingency routes and the bridge wallet	§9
T18 to T21	Pattern E — cancellations, reversals and voids	§10
T22a, T22 to T25, T24a	Pattern F — rejected payout and return leg	§11
T26, T27	Pattern H — balance conversion	§13
T28, T29	Pattern I — overpayment and sender credit	§14
T30 to T35	Pattern J — markup shares and partner rebates	§15
T36 to T38	Pattern K — variances, by kind	§16
T39, T40, T41a to T41c, T42, T43	Pattern L — loss events	§17
T44 to T51	Pattern M — the reseller project (NPL-GR)	§18
T52a, T52 to T55	Pattern N — earnings kept unconverted and offset between partners	§19
T56 to T58	Pattern O — reroute after conversion	§19a

T59	Pattern Q — month-end reconciliation true-up	§19c
L1 to L13	Acceptance vectors A, the ledger's worked deals (L10 to L13 added in 18a)	Calculation Specification §18
B1 to B23	Acceptance vectors B, NPL's own figures recomputed (B18 to B23 added in 19b)	Calculation Specification §19, numbered in 19b
W1 to W8	Guided walkthroughs through the model	Scenario Catalogue Part 3
F1 to F10	Shared operational flows	Scenario Catalogue Part 2
C1 to C10, C6a	Common procedures	Operating Procedures Part A
A.1 to A.10, D.1 to D.5	Daily rhythm, quick reference, access review; exception playbooks	Operating Procedures Parts A and D

Pattern G has no transactions: an offset is a view, not a posting. Pattern P (interruptions) has none yet; its worked postings take the next free T numbers when they are written.